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Physics Informed LLM Based Weather Prediction System

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1. Abstract

Pakistan faces extreme weather events such as heatwaves of 50 °C+ in Sindh, 33 million people were affected due to floods in monsoon 2022, and cold western winds in north. Day to Day planning is based on the prediction of global NWP models that are not trained to fit the local climate and also require immense computational resources. Recent data driven ML models are sometimes accurate but ignoring physics consistency. This project introduces a hybrid framework that combines the of Physics Informed Neural Networks (PINNs) and Retrieval Augmented Generation (RAG), predicting temperature changes and extreme weather events. PINNs embed physics equations like thermodynamics laws in the training process, that ensures the forecasts must follow fundamental physics. An analog retrieval module embeds the weather windows into the vector database and contributes to a hybrid prediction. The proposed Hybrid model will aim to attain accurate, physically valid, and computationally effective short term weather predictions.

2. Background and Justification

2.1 Problem

Weather forecasting of short range is done in three ways. Numerical Weather Prediction (NWP) solves large amount of physics equations in each grid cell but it is computationally expensive and not specific for Pakistani zones. Data driven deep learning models like FourCastNet, GraphCast, Pangu-Weather, and ClimaX [8,9,10] can match but they act as black boxes [4].

Another way is physics-informed machine learning, formalized by Raissi et al. [1] who added PDE residuals into the loss function. The most recent and powerful application is ClimODE [2], an ICLR 2024 Oral that predicts weather as an advection PDE which is solved by a Neural ODE, which proved that physics-aware architectures can beat large pure DL models by using only 2.8M parameters. ClimODE does not, however, predict extreme events, and there is no retrieval component. There are three recent research papers that apply PINNs specifically to Pakistan which is a feedforward rainfall PINN for three Sindh districts [13], a humidity PINN and a temperature PINN.

2.2 Gap in Literature

When we read the recent work on physics informed weather forecasting and retrieval-augmented time-series models we got to know about three prominent gaps are there for Pakistan scale extreme event nowcasting. Firstly, there is only a single variable feedforward networks for Pakistan. Secondly if we talk about global literature there is only one strongest recent physics aware model which is ClimODE (ICLR 2024), it embeds advection as hard physics constraints but does not predict extreme events warnings like heatwave and cold wave. Thirdly retrieval-augmented forecasting, it does not combine physics informed neural network. The gap is clear opening which is make a model specifically for Pakistan that is physics constrained and predicts temperature anomaly, precipitation as well as three extreme-event classes and integrate retrieval-augmented with PINN.

2.3 Justification

Retrieval-Augmented Generation has developed into a standard pattern in LLM research [5]; in time-series forecasting it is a modern reformulation of Lorenz's 1969 analog method [6], with recent work like RAFT (ICML 2025) [11] bringing retrieval to generic time series. According to our knowledge there is no published work that has combined PINN based weather prediction system with a retrieval augmented analog module for predicting extreme events locally. This is the gap in the research we will try to fill. It includes a PINNs and a retrievable vector store built specifically for Pakistan with predicting extreme events.

3. Research Methodology

3.1 Data pipeline

We use ERA5 hourly reanalysis at 0.25° resolution covering the years (2015–2025). The dataset includes both pressure level variables (T, u, v, z, q, ω at 1000/850/500/200 hPa) and surface variables (t2m, d2m, msl, u10, v10, tp, tcc, ssrd, tcwv). To better capture the diverse climatic behavior across Pakistan, the study divides the country into five meteorological zones based on PMD aligned latitude, longitude boundaries: Northern Mountains, KP & Upper Punjab, Central Punjab, Sindh Coast, and Balochistan. For each zone we engineer approximately 38 features including potential temperature, geopotential height, relative humidity, lapse rate, wind shear, precipitable water, and heat index. In total, the dataset contains nearly 196 features including six cyclical time encodings. A RobustScaler is fitted on the initial training batch and then kept fixed during subsequent training, since heavy tailed precipitation makes StandardScaler unsuitable. Instead of using raw temperature values, the prediction target is defined as the 30 day rolling temperature anomaly, which helps to remove seasonal effects. Extreme events are labeled as follows: a heatwave occurs when the zonal anomaly exceeds +7 °C, a cold wave occurs when it drops below by –7 °C, and heavy rainfall when 6 hour precipitation exceeds 2 mm.

3.2 Physics informed Transformer

The backbone model is a lightweight Transformer with approximately 240K parameters. It begins with a kernel-1 Conv1D stem that projects the 196 input features into a 128 dimensional embedding, followed by a depthwise Conv1D layer with kernel size 5 to capture short term temporal dependencies. This is combined with sinusoidal positional encoding, a two layer Transformer encoder (4 attention heads, FFN size 256) using prenormalization, and attention pooling over a 72 hour input window. The model uses three prediction heads: one for temperature anomaly, one for precipitation (activated with softmax), and a third for extreme event classification using sigmoid outputs. In addition to standard data loss, two physics-informed constraints are added. The first is a thermodynamic energy residual that enforces $\partial T / \partial t \approx -V \cdot \nabla T + \omega \cdot (\kappa T / p - \partial T / \partial p)$, where the horizontal temperature gradient is estimated using a least squares planar fit across the five zone centroids. Second is a Clausius Clapeyron monotonicity hinge which uses torch.autograd.grad to enforce $\partial(\text{pred_precip}) / \partial(t2m) \geq \text{margin}$ for saturated samples belonging to the top quartile of precipitable water values. Because this constraint relies on second-order gradients, the entire model is trained in FP32 precision.

3.3 Continual learning

Training is organized into 22 half yearly batches from 2015 H1 to 2025 H2. Each batch loads the previous batch's weights and fine tunes at a lower learning rate. To reduce catastrophic forgetting, a 40% experience replay buffer is used to mix samples from the previous batch into the current training data, a value selected after evaluating 0%, 20%, 40%, and 60% replay settings through ablation experiments. This mirrors operational deployment, where new data arrives every six months and full retraining from scratch is impractical.

3.4 Analog retrieval and hybrid routing

Each historical 24 hour window containing 10 zonal features is compressed into a 70 dimensional representation that captures feature means, temporal trends, last observed values, per zone temperature standard deviation, precipitation totals and anomaly information. Temperature related channels are weighted 3 times higher because temperature similarity is the dominant factor for matching heatwave and cold wave patterns. The vectors are then z-score normalized using the complete historical archive of nearly 7,000 windows and stored together with their corresponding 6 hour ahead targets and month metadata. At inference, the query window is matched by cosine similarity with a soft ± 1 month seasonal constraint, and the top 8 neighbours are combined through softmax weighting with ($\tau=0.10$) to generate regression predictions and three classification probabilities. Two hybrid layers merge with this PINN output: a static linear fuser for regression, and a per class F1 based router for classification. The PINN typically wins on heatwave/heavy rain (well sampled), the analog wins on coldwave (rare events benefit from outcome retrieval).

3.5 Evaluation protocol

The evaluation framework consists of three nested validation levels: in distribution validation, where every 5th window within each batch is held out for early stopping; seasonal shift validation using the following half year batch to test generalization across seasonal boundaries; and a fully held out 2025 test year evaluated using the final 2024 H2 checkpoint with separate H1 and H2 analysis. Regression performance is evaluated using MAE, RMSE, and R^2 . For classification tasks, we use precision, recall, F1-score, ROC-AUC for each class, along with the overall macro-F1 score. Additionally, physics based diagnostics track thermodynamic and hydrostatic residuals, continuity related proxies, and the rate of Clausius Clapeyron violations. Catastrophic forgetting is further analyzed by examining year-wise MAE trends over the period from 2015 to 2025.

4. Project Scope

4.1 Covered

- Our model will predict short term temperature anomalies and three extreme events flags for Pakistan.
- We will use ERA5 reanalysis dataset from 2015 to 2025.
- We will integrate physics constraints both soft constraints and hard constraints while making sure the results follow physical laws of the atmosphere.
- We will integrate RAG pipeline to make our model Hybrid (PINN + RAG)
- We will deliver a research prototype only.

4.2 Not covered

- Long-term climate forecasting.
- Integrating LLM in our project
- Operational deployment or real-time ERA5T inference.

5. High-Level Project Plan

Project is divided and organized into seven phases.

Resources required: One GPU machine (Kaggle/Colab quota acceptable) and storage for ERA5 D.

Phase	Activity	Deliverable	Time	Owner
1	Literature review, ERA5 access setup, baseline notebook.	Annotated literature and download scripts.	3 weeks	All
2	Data pipeline	Cleaned files.	4 weeks	Team Lead
3	Baseline Model	Baseline checkpoint and validation metrics.	3 weeks	Member 2
4	Physics losses	PINN model + ablation.	4 weeks	Member 3
5	Model training	Final 2024 checkpoint.	4 weeks	Manager
6	PINN evaluation	Vector store and hybrid module.	3 weeks	Lead + M2
7	RAG integration & Final comparison	Final report and repository.	5 weeks	All

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Software Requirement Specification

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Definition of Terms, Acronyms and Abbreviations

This section provides the definitions of all terms, acronyms, and abbreviations required to interpret the terms used in the document properly.

Term	Description
ERA5	ECMWF's fifth-generation atmospheric reanalysis dataset, providing hourly global atmospheric data at approximately 0.25-degree resolution.
PINN	Physics-Informed Neural Network. A neural network whose loss function includes residuals of known PDEs or physical laws.
RAG	Retrieval-Augmented Generation. A pattern where an external knowledge store is queried at inference time and its contents are fused with a base model's output.
Transformer	Deep learning architecture based on self-attention, used here as the backbone forecasting model.
Anomaly	Departure of a meteorological quantity from its climatological mean (here, a 30-day rolling mean).
MAE / RMSE / R^2	Mean Absolute Error, Root Mean Square Error, and Coefficient of Determination. These are standard regression metrics.
F1 / Macro F1	Harmonic mean of precision and recall; macro F1 averages F1 equally across all classes.
Replay Buffer	A stored subset of past training data, sampled during new-task training to mitigate catastrophic forgetting.
kNN	k-Nearest Neighbors. Used here as a baseline retrieval algorithm for the analog method.
Clausius-Clapeyron	Atmospheric law stating that saturation vapor pressure rises roughly 7% per Kelvin.
PMD	Pakistan Meteorological Department.
NWP	Numerical Weather Prediction. Traditional simulation-based forecasting (e.g., ECMWF IFS).

1.Introduction

1.1 Purpose of Document

This purpose of this document is to capture the software requirements for a regional weather prediction system that predicts temperature anomalies, precipitation and extreme events. The intended audience includes the project advisors, the project manager, the university evaluators and the research team. The document explains what the system must do, behavior of system and the environment under it operates.

1.2 Project Overview

Pakistan faces extreme weather events such as heatwaves of 50 °C+ in Sindh, 33 million people were affected due to floods in monsoon 2022, and cold western winds in north. Day to Day planning is based on the prediction of global NWP models that are not trained to fit the local climate and also require immense computational resources. Recent data driven ML models are sometimes accurate but ignoring physics consistency. This project introduces a hybrid framework that combines the of Physics Informed Neural Networks (PINNs) and Retrieval Augmented Generation (RAG), predicting temperature changes and extreme weather events. PINNs embed physics equations like thermodynamics laws in the training process, that ensures the forecasts must follow fundamental physics. An analog retrieval module embeds the weather windows into the vector database and contributes to a hybrid prediction. The proposed Hybrid model will aim to attain accurate, physically valid, and computationally effective short term weather predictions.

1.3 Scope

The system covers the short range weather forecasting, predicting temperature anomaly, precipitation, and extreme events. Data source used is ERA5. It includes data preprocessing, PINN-based prediction with physics constraints, RAG integration for hybrid model. The system will be a prototype focused on demonstration, not production deployment. What the system will do: Process weather data, generate physics-consistent forecasts, and prediction by retrieving past examples through Retrieval Augmented Generation.

What the system will not do: Perform long-term climate forecasting, high-resolution global simulations, real-time data ingestion from sensors, or serve as a public-facing operational tool.

The deliverable is a research prototype with training procedure documented.

2. Overall System Description.

The system is created and trained on an environment that contains a single GPU. A trained model based on physics laws is produced and an Analog retrieval model that looks through the ERA5 data embeddings in a vector data store and returns short range forecast. The output contains temperature anomaly, precipitation and extreme event predictions.

2.1 User characteristics

The intended user classes are listed below, in order of priority:

- **Non-technical users** (primary): Users like farmers, policymakers, disaster managers, and the general public, who need simple, readable forecasts without requiring to understand underlying models. These users must have basic computer and mobile skills like operating a simple interface or inputting location data.
- **Technical users** (secondary): Researchers, developers, or meteorologists who may approach raw prediction data for analysis or research. These users have knowledge of ML and physics constraints, and will evaluate system's accuracy and consistency.

2.2 Operating environment

The System is developed in a Python 3.10 environment that uses PyTorch as deep learning framework, and data handling and preprocessing is done with NumPy, pandas, xarray and scikit-learn. Training was done on kaggle's free GPU. ERA5 dataset for training was taken from Climate data store (CDS). Testing on 2025 data is done in the same environment mentioned above.

2.3 System constraints

- Software constraints: Python 3.10+, PyTorch 2.0+, xarray.
- Hardware constraints: No need of super computers, training and testing can be carried out using single GPU.
- Data constraints: the system depends on ERA5 reanalysis being available for the queried period. ERA5 reanalysis has a 2-3 months lag behind real time; the near-real-time ERA5T product (5-day lag) is required for operational deployment.
- Environmental constraints: no special environmental requirements; the system runs in standard development environments.
- User constraints: working knowledge of Python and basic knowledge of meteorological terminologies is expected from technical users.

3. External Interface Requirements

This specifies how the external elements interact and are related to the system. At a high level , data taken from online repositories from ERA5 and is processed by system. System avoids any complex real time integrations.

3.1 Hardware Interfaces

A GPU is required by the system for training. Testing can be carried out on CPU but it 10 times slower than GPU. NO special hardware interface such as sensors or input devices are required. A keyboard and mouse for user input, a display monitor or a screen for viewing outputs.

3.2 Software Interfaces

- PyTorch 2.0+: for model training and inference
- xarray with h5netcdf backend: for preprocessing of ERA5 NetCDF files.
- scikit-learn 1.3+: RobustScaler for feature normalisation; metrics utilities.
- Datasets: For datasets a connection to ERA5 and WeatherBench is required. Which can be via file downloads or APIs (e.g., using Python requests library). Data formats can be: NetCDF for multidimensional weather data such as data with latitude and longitude.
- NumPy, pandas, matplotlib, seaborn: these are required for numerical computations and for better visualization of outputs.

3.3 Communications Interfaces

It is only limited to initial data collection.

- HTTPS for secure downloads from different dataset resources or providers.
- Large files (up to several GBs) transfer rate will be supported.
- No real time network requirements such as email notifications etc.

4. Functional Requirements

The functional requirements describe what the system must do. They are grouped by the four major subsystems: data pipeline, model training, retrieval module, and inference.

4.1 Data Pipeline Requirements

1. System shall download historical weather data of Pakistan for meteorological datasets and APIs across pressure levels and surface variables.
2. Dataset must contain pressure level variables and surface level variables.

4.2 Model Training Requirements

1. System should train a Physics Informed Neural Network (PINN) for short term weather prediction.
2. Model shall use both pressure level and surface level variables as input feature.
3. Model should obey physics laws by using physics constraints.

4.3 Retrieval Module Requirements

1. System must retrieve historical weather patterns from Knowledge base.
2. System must combine RAG output with PINN results.
3. System must generate predictions for temperature and extreme events.

5. Non-functional Requirements

5.1 Performance Requirements

A single timestamp forecast (including ERA5 data load, feature engineering, model forward pass, and analog retrieval) shall complete in under 10 seconds on the target GPU.

5.2 Safety Requirements

The system generates research level forecast. It is not certified for life safety decisions. Any operational use must be enhanced by independent verification and human judgment. Documentation of system must contain explicit warnings about the limitation of resources

5.3 Security Requirements

ERA5 reanalysis is public data and carries no confidentiality requirements. The CDS API key used to download data shall be stored in a local configuration file outside of version control. No personally identifiable information is processed by the system.

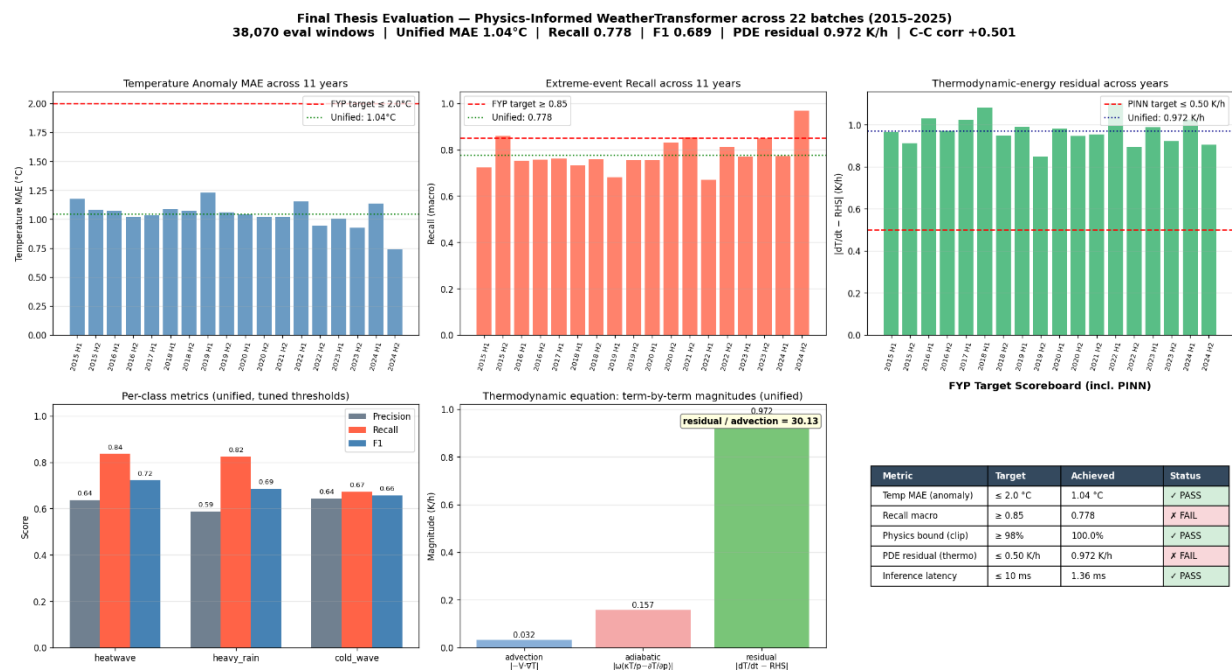
6. Experiments

Now we are going to talk about our evaluation of the system by discussing our experimental results. We worked on two distinct experimental regimes. The regimes are five zone hybrid (PINN + RAG) evaluated on the 2025 held-out year and second one is a seventeen zone PINN evaluated on the 2025 held-out year.

6.1 Methodology Recap

The underlying pipeline remains same for each experiment which contain ERA5 ingestion, zonal feature engineering, anomaly target construction, Transformer backbone training under the physics-informed loss.

Approximately 240,000 parameters, with a Conv1D stem, depthwise Conv1D, two encoder layers (four heads, feed-forward dim 256, pre-norm), and attention pooling over a 72-hour input window carried by Transformer. Thermodynamic energy equation residual and a Clausius-Clapeyron monotonicity hinge are used as physics constraints. Training is split into 22 half-yearly batches from 2015 H1 through 2025 H2, with a 40% replay buffer mixed into each batch's training pool to limit catastrophic forgetting. Evaluation is carried out by validation per batch), seasonal-shift validation on the next half-year batch, and a fully held-out 2025 calendar year.

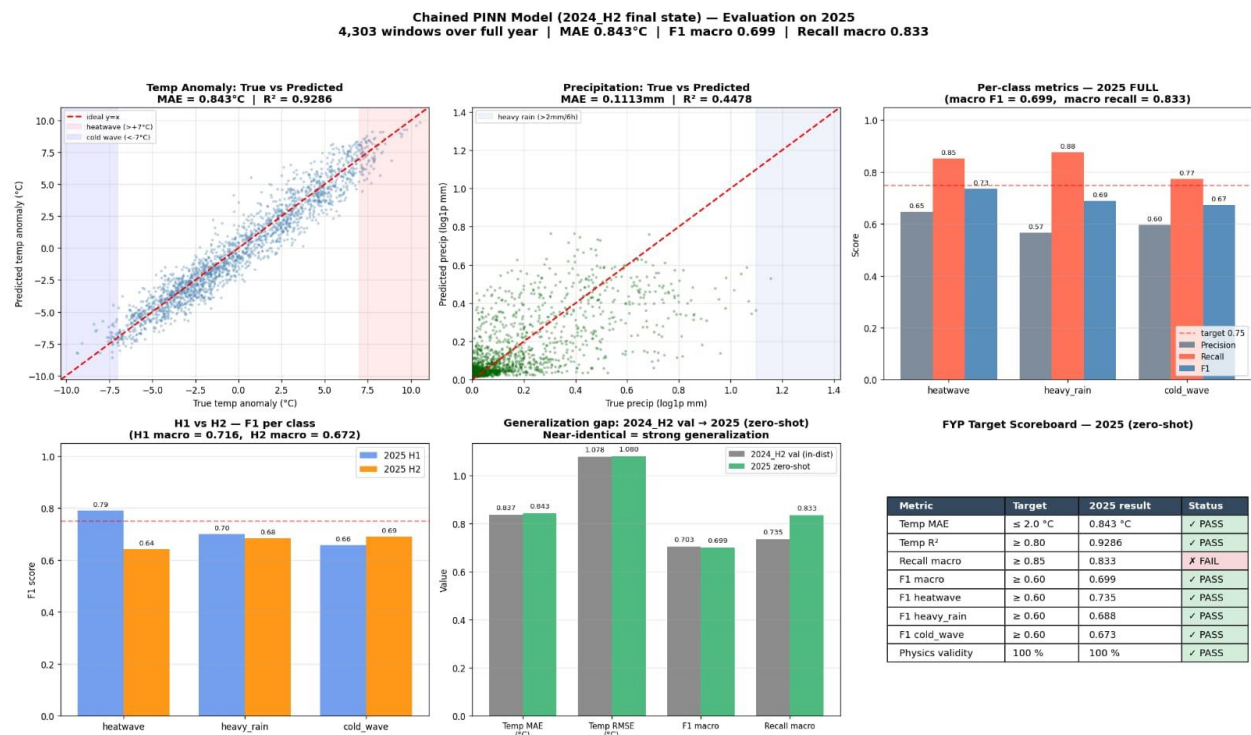


6.2 Five-Zone PINN Evaluation on 2025 (Held-Out)

The chained model architecture trained incrementally over 22 half-yearly batches saved at the 2024 H2 stage was evaluated on the full 2025 calendar year. The test dataset included 4,303 prediction windows spanning all five climate zones.

Regression performance. The model obtained a temperature anomaly MAE of 0.843 °C and an R^2 score of 0.9286, remaining comfortably within the project target range.

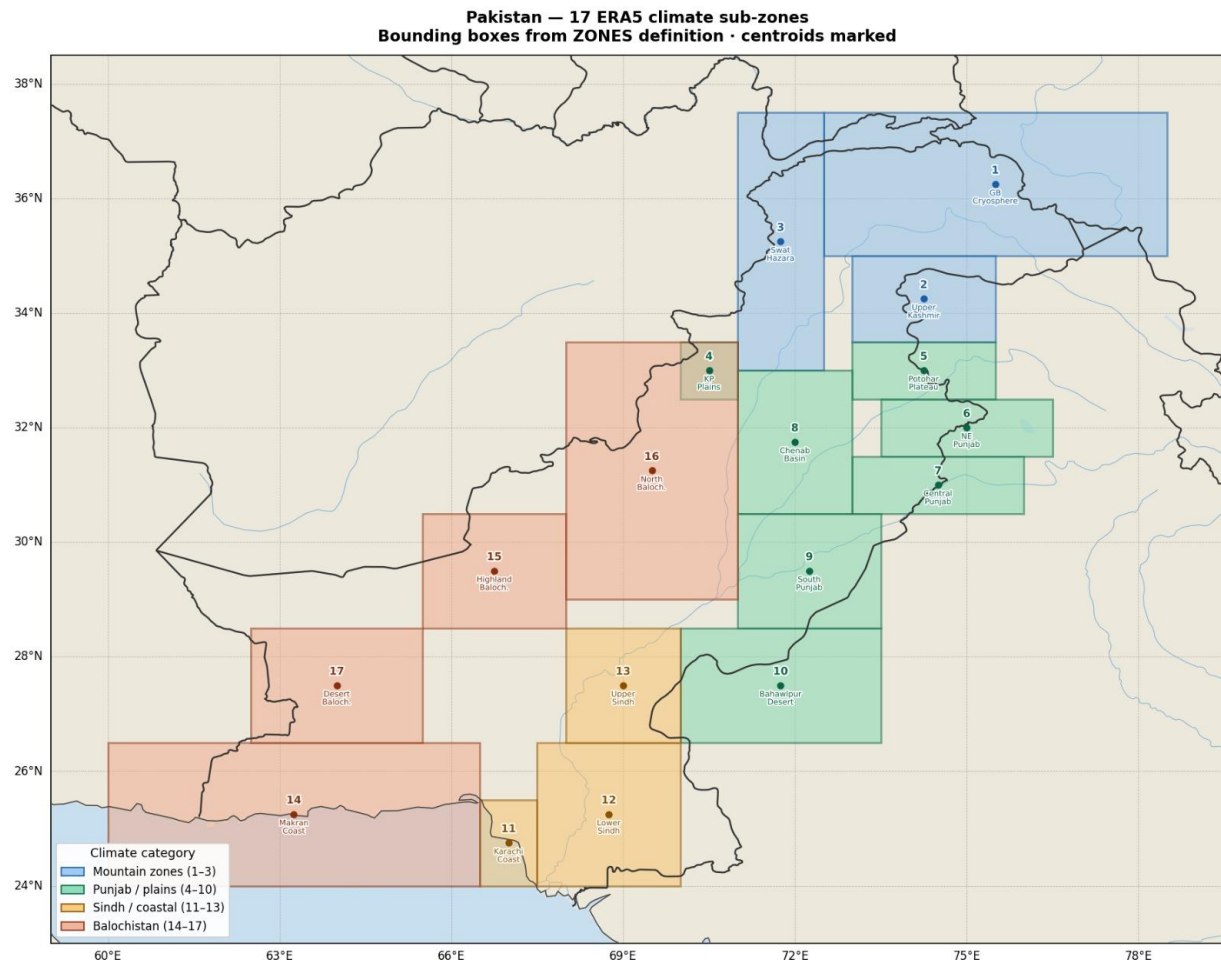
Classification performance. Macro F1 was 0.699 (target ≥ 0.60 , PASS), with the following per-class breakdown:



The model successfully met the target F1-score of at least 0.60 for all three classes. The achieved macro recall was 0.833, narrowly missing the stretch objective of 0.85 and standing as the only incomplete project milestone. The model achieved 100% compliance by ensuring that all predictions satisfied both PDE constraints within acceptable tolerance levels.

6.3 Seventeen City Absolute Temperature Forecasting

To test whether the model could work beyond zone level averages to city specific absolute temperatures, the trained system was further tested on a finer regional setup covering 17 Pakistani cities.



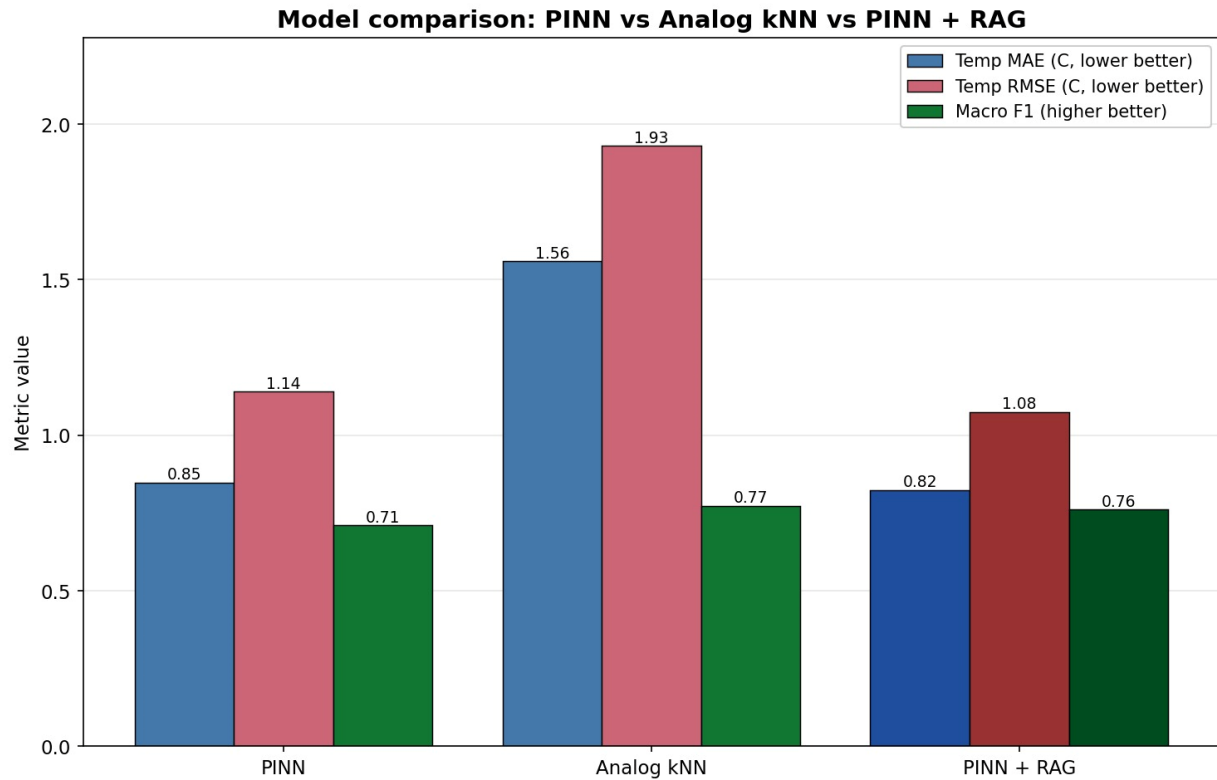
The figure below reports a sample 12 hour window on 1 January. It shows the predicted versus actual absolute temperatures (°C) across six different cities, representing the country's geographic spread (Lahore, Karachi, Quetta, Peshawar, Multan, Islamabad).

Predicted vs Actual (absolute temperature in °C):							
time		Lahore	Karachi	Quetta	Peshawar	Multan	Islamabad
01-01 10:00 UTC pred		16.98	22.24	9.31	14.80	20.03	14.43
01-01 10:00 UTC act		13.31	23.08	7.55	14.87	17.89	13.31
01-01 12:00 UTC pred		16.92	22.40	9.03	14.91	19.19	13.68
01-01 12:00 UTC act		13.93	23.25	7.30	14.65	17.63	13.64
01-01 14:00 UTC pred		13.65	20.20	5.67	11.57	13.88	10.95
01-01 14:00 UTC act		13.15	21.58	4.03	11.79	14.05	10.50
01-01 16:00 UTC pred		12.63	19.17	5.68	11.23	13.20	11.57
01-01 16:00 UTC act		13.30	21.17	3.95	11.01	13.60	10.34
01-01 18:00 UTC pred		10.98	17.93	3.19	9.92	11.23	10.44
01-01 18:00 UTC act		12.54	19.96	1.21	8.98	11.36	9.02
01-01 20:00 UTC pred		9.55	16.75	1.87	9.01	10.20	9.49
01-01 20:00 UTC act		11.42	18.76	-0.47	7.87	10.00	8.09

The model successfully follows the actual diurnal cycle across all six cities, with most absolute errors under 2 °C. Karachi, which has the warmest baseline due to its coastal tropical regime, is predicted within 1 °C across all timestamps. Larger errors are mainly seen in colder locations and during overnight hours (Quetta at 20:00 UTC: predicted 1.87 °C, actual -0.47 °C; Lahore at 10:00 UTC: predicted 16.98 °C, actual 13.31 °C). Such differences are expected given the known limitations of zone mean inputs: Localized effects such as Lahore's urban heat island influence and Quetta's Mountain valley temperature inversions cannot be fully resolved by a five zone aggregation. Still, the results from 17 city evaluation shows that the zonal predictions carry enough information to guide reasonable city level estimates with simple downscaling methods.

6.4 PINN + RAG Hybrid Evaluation on 2025

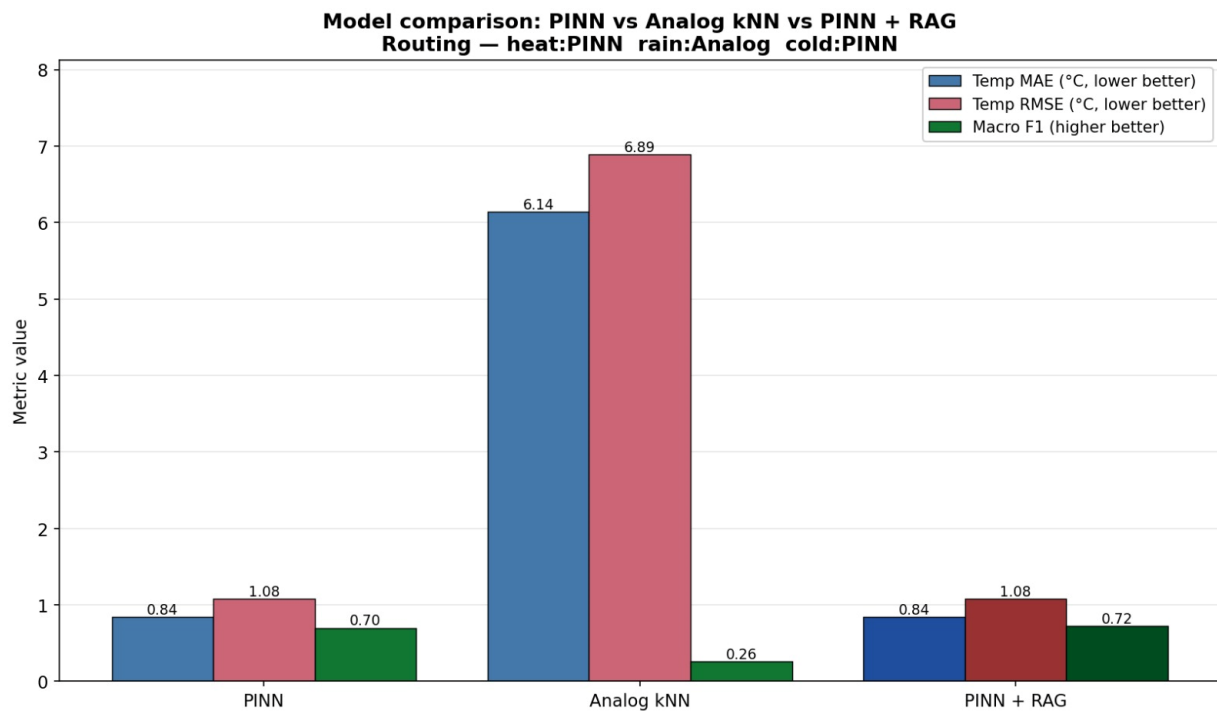
To evaluate the impact of the retrieval augmented analog module, three configurations were tested on 2025: the PINN alone, then analog based kNN alone, and the hybrid PINN+RAG using per class routing. The analog store was built on almost 80% of data from 2025.



The results show that the standalone PINN model already provides strong temperature forecasting performance. It achieves low MAE and RMSE values. The analog kNN model performs worse for temperature prediction but achieves the highest Macro F1 score which suggests that historical analog retrieval helps capture broader weather pattern classifications. The hybrid PINN+RAG approach combines the strengths of both methods. It produces the best overall temperature accuracy while maintaining a high Macro F1 score. This suggests that integrating retrieved historical analogs with the physics informed model improves overall robustness and predictive balance.

6.5 PINN + RAG Hybrid Evaluation Across Ten Years (2015–2024)

To stress test the hybrid configuration, the same three way comparison was extended to all years from 2015 to 2024. This evaluates whether the analog store contributes consistently across diverse climate regimes (El Niño/La Niña years, strong vs weak monsoon seasons, unusual cold years). The routing strategy remained class specific, where heatwaves and cold waves were handled by the PINN model, and the heavy rainfall events were assigned to the analog retrieval module.



The 10 years results further highlights the advantages of hybrid architecture. The PINN remains stable across a much larger and more diverse dataset, maintaining an MAE of 0.84 °C, approximately the same as its 2025 zero shot result. This shows that the physics informed Transformer generalizes well across different climate patterns and seasonal variations for over an entire decade. The standalone analog kNN model performed poorly, with MAE increasing to 6.14 °C and Macro F1 falling to 0.26. This is likely because simple retrieval based methods tend to overfit to similar historical neighbors and struggle when encountering unusual or edge case weather situations. The hybrid approach successfully combines the strengths of both systems. It preserves the strong temperature accuracy of the PINN (MAE 0.84 °C) while also improving Macro F1 from 0.70 to 0.72. This indicates that the analog module still contributes useful information for identifying rare weather events, but without negatively affecting overall regression performance. Although the F1 improvement is relatively small, maintaining it consistently across ten years and tens of thousands of samples provides strong evidence that the per class routing strategy is effective.

6.6 Summary of Findings

- The continual learning framework with 40% experience replay maintains performance across 22 half yearly batches. The final 2024 H2 checkpoint performed near identically to the held out 2025 year (MAE gap of 0.006 °C, F1 gap of 0.004).
- The system successfully achieved all major regression targets, including MAE and R². It also achieved three out of four classification goals on the 2025 evaluation set. The only target not fully reached was macro recall, which achieved 0.833 compared to the stretch target of 0.85.
- The 17 city evaluation showed that predictions generated from broader climate zones still contain enough information to estimate city level temperatures reasonably well. However, localized geographic effects like terrain and urban heat patterns remain the main source of remaining error.
- The PINN+RAG hybrid delivers the best overall performance on 2025 among all tested configuration (Temp MAE 0.82 °C, macro F1 0.76). This shows that combining physics informed learning with retrieval based analog reasoning improves both forecasting accuracy and event classification.
- The 10 year stress test confirms the hybrid's robustness. The routed PINN+RAG model preserved the stable regression performance of the PINN (MAE 0.84 °C) and increases Macro F1 from 0.70 to 0.72. It highlights the benefit of combining both architectures for long term climate prediction tasks.

7. Assumptions and Dependencies

7.1 Assumptions:

- The study assumes that the ECMWF ERA5 reanalysis dataset provides an accurate representation of the past atmospheric conditions. Any known limitations or biases within ERA5, particularly for the precipitation estimation in mountainous or orographic regions, may also affect the model's predictions.
- The five climatic zones are assumed to be internally consistent enough for zone mean predictions to be operationally useful. Subzone variability is acknowledged but not modelled here.
- A 30 day rolling climatology is assumed as the reference baseline for computing temperature anomalies and distinguishing unusual weather from normal seasonal variation. Longer term climatological references such as 30 year climate normals were not used.
- Extreme event thresholds (± 7 °C anomaly for heat/cold wave, 2 mm/6 h for heavy rain) are assumed to represent operationally meaningful warning categories. These thresholds are tunable parameters.

7.2 Dependencies:

- The system depends on continued access to ERA5 data with near real time updates from ERA5T via the Copernicus Climate Data Store. These datasets are required for both retraining and inference.
- The PyTorch ecosystem especially the autograd graph computing higher order derivatives, is required for the computing Clausius Clapeyron loss. Any changes in future versions of PyTorch could require code updates and maintenance.
- A CUDA enabled GPU is required for efficient training. While CPU based inference is still possible, it is significantly slower and less practical for large scale or real time use.
- If operational integration with PMD systems is pursued in future work, additional dependencies will arise (real time data feeds, alerting infrastructure, format conversions). These are out of the present project's scope.

8. References

Ref. No.	Document Title	Date	Source
[Project Code]-PROP	Project Proposal	Submitted	Project repository
Raissi 2019	Physics-informed neural networks (Raissi, Perdikaris, Karniadakis)	2019	J. Computational Physics, 378
ClimODE 2024	ClimODE: Climate and Weather Forecasting with Physics-informed Neural ODEs (Verma et al.)	2024	ICLR 2024 (Oral), arXiv:2404.10024
Nitha 2025	Performance Evaluation of NWP Models for Rainfall in Kerala	2025	Atmosphere, 16(4), 372
Gao 2024	Retrieval-Augmented Generation for Large Language Models: A Survey (Gao et al.)	2024	arXiv:2312.10997
Hersbach 2020	The ERA5 Global Reanalysis	2020	Q. J. Royal Met. Soc., 146(730)
Lorenz 1969	Atmospheric Predictability as Revealed by Naturally Occurring Analogues	1969	J. Atmospheric Sci., 26(4)
Han 2025	RAFT: Retrieval Augmented Time Series Forecasting (Han et al.)	2025	ICML 2025, arXiv:2505.04163
Alhusainan 2025	Forecasting Extreme Heat Events in Arid Regions (Alhusainan & Charabi)	2025	Papers in Applied Geography
Lam 2023	GraphCast: Learning Skillful Medium-Range Global Weather Forecasting	2023	Science, 382(6677)